MSG No. 38.193 A_bmm2 [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `mm2`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4]`
2	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,6,7,8]`
3	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[9,10,11,12]`
4	$\left[0,\ \frac{1}{2},\ z\right]$	`[13,14,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `mm'2'`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ z\right]$	`[1,3,14,16]`
2	$\left[0,\ \frac{3}{4},\ z\right]$	`[2,4,13,15]`
3	$\left[0,\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[5,7,10,12]`
4	$\left[0,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[6,8,9,11]`

Table 3: Wyckoff site: `4c`, site symmetry: `mm2`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[13,14,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `mm'2'`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{4},\ z\right]$	`[1,3,14,16]`
2	$\left[\frac{1}{2},\ \frac{3}{4},\ z\right]$	`[2,4,13,15]`
3	$\left[\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[5,7,10,12]`
4	$\left[\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[6,8,9,11]`

Table 5: Wyckoff site: `8e`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,4]`
2	$[-x,\ 0,\ z]$	`[2,3]`
3	$\left[x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,8]`
4	$\left[-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[6,7]`
5	$\left[x,\ 0,\ z+\frac{1}{2}\right]$	`[9,12]`
6	$\left[-x,\ 0,\ z+\frac{1}{2}\right]$	`[10,11]`
7	$\left[x,\ \frac{1}{2},\ z\right]$	`[13,16]`

8	$\left[-x,\, \frac{1}{2},\, z\right]$	[14,15]

Table 6: Wyckoff site: 8f, site symmetry: .m'.

No.	position	mapping
1	$\left[x,\, \frac{1}{4},\, z\right]$	[1,16]
2	$\left[-x,\, \frac{3}{4},\, z\right]$	[2,15]
3	$\left[-x,\, \frac{1}{4},\, z\right]$	[3,14]
4	$\left[x,\, \frac{3}{4},\, z\right]$	[4,13]
5	$\left[x,\, \frac{3}{4},\, z+\frac{1}{2}\right]$	[5,12]
6	$\left[-x,\, \frac{1}{4},\, z+\frac{1}{2}\right]$	[6,11]
7	$\left[-x,\, \frac{3}{4},\, z+\frac{1}{2}\right]$	[7,10]
8	$\left[x,\, \frac{1}{4},\, z+\frac{1}{2}\right]$	[8,9]

Table 7: Wyckoff site: 8g, site symmetry: m..

No.	position	mapping
1	$[0,\, y,\, z]$	[1,3]
2	$[0,\, -y,\, z]$	[2,4]
3	$\left[0,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[5,7]
4	$\left[0,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[6,8]
5	$\left[0,\, y,\, z+\frac{1}{2}\right]$	[9,11]
6	$\left[0,\, -y,\, z+\frac{1}{2}\right]$	[10,12]
7	$\left[0,\, y+\frac{1}{2},\, z\right]$	[13,15]
8	$\left[0,\, \frac{1}{2}-y,\, z\right]$	[14,16]

Table 8: Wyckoff site: 8h, site symmetry: m..

No.	position	mapping
1	$\left[\frac{1}{2},\, y,\, z\right]$	[1,3]
2	$\left[\frac{1}{2},\, -y,\, z\right]$	[2,4]
3	$\left[\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[5,7]
4	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[6,8]
5	$\left[\frac{1}{2},\, y,\, z+\frac{1}{2}\right]$	[9,11]
6	$\left[\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	[10,12]
7	$\left[\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[13,15]
8	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[14,16]

Table 9: Wyckoff site: 16i, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-x,\, -y,\, z]$	[2]
3	$[-x,\, y,\, z]$	[3]

continued ...

Table 9

No.	position	mapping
4	$[x,\, -y,\, z]$	[4]
5	$\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[5]
6	$\left[-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[6]
7	$\left[-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[7]
8	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[8]
9	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[9]
10	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[10]
11	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[11]
12	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[x,\, y+\frac{1}{2},\, z\right]$	[13]
14	$\left[-x,\, \frac{1}{2}-y,\, z\right]$	[14]
15	$\left[-x,\, y+\frac{1}{2},\, z\right]$	[15]
16	$\left[x,\, \frac{1}{2}-y,\, z\right]$	[16]